arXiv:1409.6411v1 [math.CA] 23 Sep 2014

SEVERAL COMPLETELY MONOTONE FUNCTIONS RELATED TO DETEMPLE'S SEQUENCE

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Abstract. In this paper, we present the necessary and sufficient conditions such that several functions involving $R(x)=\psi(x+1/2)-\ln x$ with a parameter are completely monotone on $(0,\infty)$, where ψ is the digamma function. This generalizes some known results and verifies a conjecture posed by Chen.

1. Introduction

A function f is said to be completely monotonic on an interval I , if f has derivatives of all orders on I and satisfies

$$(-1)^n f^{(n)}(x)\geq 0 \text{ for all } x\in I \text{ and } n=0,1,2,.... \tag{1.1}$$

If the inequality (1.1) is strict, then f is said to be strictly completely monotonic on I. It is known (Bernstein's Theorem) that f is completely monotonic on $(0,\infty)$ if and only if

$$f(x)=\int_0^\infty e^{-xt}d\mu(t),$$

where μ is a nonnegative measure on $[0,\infty)$ such that the integral converges for all $x>0$, see [1, p. 161].

For $x>0$ the classical Euler's gamma function Γ and psi (digamma) function ψ are defined by

$$\Gamma(x)=\int_0^\infty t^{x-1}e^{-t}dt, \qquad \psi(x)=\frac{\Gamma'(x)}{\Gamma(x)}, \tag{1.2}$$

respectively. The derivatives ψ', ψ'', ψ''', ... are known as polygamma functions.

As the important role played in many branches, such as mathematical physics, probability, statistics, engineering, the gamma and polygamma functions have attracted the attention of many scholars. In particular many authors published numerous interesting inequalities for the Euler-Mascheroni constant defined by

$$\gamma=\lim_{n\to\infty}D_n:=\lim_{n\to\infty}\left(\sum_{k=1}^{n}\frac{1}{k}-\ln n\right)=0.577215664...,$$

as well as digamma (psi) function ψ due to $\psi(n+1)=H_n-\gamma$.

Date: July 15, 2014.

2010 *Mathematics Subject Classification.* 33B15, 26D15.

Key words and phrases. Psi function, completely monotone function, inequality.

This paper is in final form and no version of it will be submitted for publication elsewhere.

It is known that the sequence D_n converges very slowly to its limit, due to

$$\frac{1}{2\left(n+1\right)}<D_n-\gamma<\frac{1}{2n}$$

(see [2], [3]). Some quicker approximations to the Euler-Mascheroni constant were established in [4], [5], [6], [7], [8], [9], [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27].

In 1993, DeTemple [7] defined a modified sequence

$$R_n=\sum_{k=1}^{n}\frac{1}{k}-\ln\left(n+\frac{1}{2}\right) \tag{1.3}$$

to approximate to γ, which converges faster since

$$\frac{1}{24\left(n+1\right)^2}<R_n-\gamma<\frac{1}{24n^2}, \tag{1.4}$$

and

$$\frac{7}{960}\frac{1}{\left(n+1\right)^4}<R_n-\gamma-\frac{1}{24n^2}<\frac{7}{960}\frac{1}{n^4}. \tag{1.5}$$

Villarino [14, Theorem 1.7] proved that for $n\in\mathbb{N}$, the following inequality

$$\tfrac{1}{24(n+1/2)^2+21/5}<R_n-\gamma<\tfrac{1}{24(n+1/2)^2+1/(1-\ln 3+\ln 2-\gamma)-54} \tag{1.6}$$

is valid with the best constants $21/5$ and $1/\left(1-\ln 3+\ln 2+\operatorname{Psi}\left(1\right)\right)-54\approx 3.7393$. Chen [18] proved that

$$\tfrac{1}{24}\left(n+\lambda\right)^{-2}<R_n-\gamma<\frac{1}{24}\left(n+\frac{1}{2}\right)^{-2}, \tag{1.7}$$

where

$$\lambda=\frac{1}{2\sqrt{6\left(1+\operatorname{Psi}\left(1\right)-\ln 3+\ln 2\right)}}-1\approx 0.551\,07$$

and $1/2$ are the best constants.

Mortici [28, Theorem 2.1] gave new bounds as follows:

$$\frac{1}{24}\left(n+\frac{1}{2}+\frac{7}{80n}\right)^{-2}<R_n-\gamma<\frac{1}{24}\left(n+\frac{1}{2}\right)^{-2}. \tag{1.8}$$

In [29], [30] (also see [31]), it was established that

$$\gamma+\ln\left(n+\tfrac{1}{2}\right)<\sum_{k=1}^{n}\frac{1}{k}\leq\gamma+\ln\left(n+e^{1-\gamma}-1\right),$$

which is equivalent to

$$0<R_n-\gamma\leq\ln\frac{n+e^{1-\gamma}-1}{n+1/2}.$$

On the other hand, Karatsuba [32] proved that for all integers $n\geq 1$, $H(n)<H(n+1)$, where $H(n)$ is defined by

$$H(n)=\left(R_n-\gamma\right)n^2. \tag{1.9}$$

In view of $\psi\left(n+1\right)=H_n-\gamma$, we see that

$$H(n)=\left(R_n-\gamma\right)n^2=\left(\psi\left(n+1\right)-\ln\left(n+\frac{1}{2}\right)\right)n^2.$$

As mentioned in [33], some computer experiments also seem to indicate that $(1+1/n)^2 H(n)$ is a decreasing convex function. Chen [15, Theorem 2] proved that for all integers $n \geq 1, H(n)$ and $((n+1/2)/n)^2 H(n)$ are both strictly increasing concave sequences, while $[((n+1)/n)^2 H(n)$ is strictly decreasing convex sequence. Also, he conjectured in [15, Cojecture 1] that

(i) the functions $H(x)$ and $((x+1/2)/x)^2 H(x)$ are both so-called Bernstein function on $(0,\infty)$. That is,

$$\begin{aligned} H(x) &> 0, \quad (-1)^n [H(x)]^{(n+1)} > 0, \\ ((x+1/2)/x)^2 H(x) &> 0, \quad (-1)^n \left[((x+1/2)/x)^2 H(x)\right]^{(n+1)} > 0 \end{aligned}$$

hold for $x > 0$ and $n \in \mathbb{N}$.

(ii) The function $((x+1)/x)^2 H(x)$ is strictly completely monotonic on $(0,\infty)$.

We remark that the functions $H(x)$ is not Bernstein function on $(0,\infty)$ due to

$$\begin{aligned} -H''(0^+) &= 2\ln 2 - 2\gamma \approx -0.231\,86 < 0, \\ -H''(1/2) &= 2\gamma + 4\ln 2 + \frac{7}{2}\zeta(3) - \pi^2 + \frac{7}{4} \approx 0.014615 > 0. \end{aligned}$$

The aim of this paper is to prove the complete monotonicity of certain functions involving

$$R(x) = \psi(x+1/2) - \ln x. \tag{1.10}$$

In Section 2, some useful lemmas are given. Our main results are presented in Section 3, in which Theorem 1 indicates that the second conjecture posed by Chen is valid, while Theorems 2–4 show the necessary and sufficient conditions such that three functions involving $R(x)$ with a parameter are completely monotone on $(0,\infty)$. In the last section, some remarks, conjecture and open problem are presented.

2. Lemmas

In order to prove our results, we need some lemmas.

Lemma 1. *For $t > 0$, let the function Q defined by*

$$Q(t) = \frac{1}{t} - \frac{1}{2\sinh\frac{t}{2}}. \tag{2.1}$$

Then

$$R(x) = \int_0^\infty e^{-xt} Q(t)\,dt, \tag{2.2}$$

$$xR(x) = \int_0^\infty e^{-xt} Q'(t)\,dt, \tag{2.3}$$

$$x^2 R(x) = \frac{1}{24} + \int_0^\infty e^{-xt} Q''(t)\,dt, \tag{2.4}$$

$$x^3 R(x) = \frac{1}{24}x + \int_0^\infty e^{-xt} Q'''(t)\,dt, \tag{2.5}$$

$$x^4 R(x) = \frac{1}{24}x^2 - \frac{7}{960} + \int_0^\infty e^{-xt} Q^{(4)}(t)\,dt. \tag{2.6}$$

Proof. Using the integral representations [34, p. 259]

$$\psi(x)=\int_0^\infty\left(\frac{e^{-t}}{t}-\frac{e^{-xt}}{1-e^{-t}}\right)dt \quad \text{and} \quad \ln x=\int_0^\infty\frac{e^{-t}-e^{-xt}}{t}dt$$

gives

$$\begin{aligned} R(x) &= \psi(x+\tfrac{1}{2})-\ln x=\int_0^\infty\left(\frac{e^{-xt}}{t}-\frac{e^{-(x+1/2)t}}{1-e^{-t}}\right)dt \\ &= \int_0^\infty e^{-xt}\left(\frac{1}{t}-\frac{1}{2\sinh\frac{t}{2}}\right)dt=\int_0^\infty e^{-xt}Q(t)\,dt. \end{aligned}$$

An integration by part yields

$$\begin{aligned} xR(x) &= x\int_0^\infty e^{-xt}Q(t)\,dt=-\int_0^\infty Q(t)\,de^{-xt} \\ &= -e^{-xt}Q(t)\,|_0^\infty+\int_0^\infty e^{-xt}Q'(t)\,dt=\int_0^\infty e^{-xt}Q'(t)\,dt, \end{aligned}$$

where the last equality holds due to $\lim_{t\to\infty}\left(e^{-xt}Q(t)\right)=\lim_{t\to0}\left(e^{-xt}Q(t)\right)=0$.

Integrations by parts again and noticing that

$$\lim_{t\to\infty}\left(e^{-xt}Q'(t)\right)=0 \ \text{ and } \ \lim_{t\to0}\left(e^{-xt}Q'(t)\right)=-\frac{1}{24}$$

lead to

$$\begin{aligned} x^2R(x) &= x\int_0^\infty e^{-xt}Q'(t)\,dt=-e^{-xt}Q'(t)\,|_0^\infty+\int_0^\infty e^{-xt}Q''(t)\,dt \\ &= \frac{1}{24}+\int_0^\infty e^{-xt}Q''(t)\,dt. \end{aligned}$$

Similarly, we have

$$\begin{aligned} x^3R(x) &= \frac{1}{24}x+x\int_0^\infty e^{-xt}Q''(t)\,dt=\frac{1}{24}x-e^{-xt}Q''(t)\,|_0^\infty+\int_0^\infty e^{-xt}Q'''(t)\,dt \\ &= \frac{1}{24}x+\int_0^\infty e^{-xt}Q'''(t)\,dt \end{aligned}$$

due to $\lim_{t\to\infty}\left(e^{-xt}Q''(t)\right)=\lim_{t\to0}\left(e^{-xt}Q''(t)\right)=0$ and

$$\begin{aligned} x^4R(x) &= \frac{1}{24}x^2+x\int_0^\infty e^{-xt}Q'''(t)\,dt=\frac{1}{24}x^2-e^{-xt}Q'''(t)\,|_0^\infty+\int_0^\infty e^{-xt}Q^{(4)}(t)\,dt \\ &= \frac{1}{24}x^2-\frac{7}{960}+\int_0^\infty e^{-xt}Q^{(4)}(t)\,dt \end{aligned}$$

due to

$$\lim_{t\to\infty}\left(e^{-xt}Q'''(t)\right)=0 \ \text{ and } \ \lim_{t\to0}\left(e^{-xt}Q'''(t)\right)=-\frac{7}{960}.$$

This completes the proof. □

In the same method as Lemma 5 in [35], we can prove the following statement.

Lemma 2 ([35, Lemma 5])**.** *Let $P(t)$ be a power series which is convergent on $(0,\infty)$ defined by*

$$P(t)=\sum_{i=m+1}^{\infty}a_it^i-\sum_{i=0}^{m}a_it^i,$$

where $a_i \geq 0$ *for* $i \geq m+1$ *with* $\max_{i\geq m+1}(a_i) > 0$ *and* $a_m > 0$, $a_i \geq 0$ *for* $0 \leq i \leq m-1$. *Then there is a unique number* $t_0 \in (0,\infty)$ *to satisfy* $P(t) = 0$ *such that* $P(t) < 0$ *for* $t \in (0, t_0)$ *and* $P(t) > 0$ *for* $t \in (t_0, \infty)$.

Using Lemma 2 we can prove the following

Lemma 3. *Let the function* Q *defined by (2.1). Then* $Q'(t) \geq c_0 Q(t)$ *for* $t > 0$, *where*

$$c_0 = \min_{t>0}\left(\frac{Q'(t)}{Q(t)}\right) \approx -0.061875. \tag{2.7}$$

Proof. Differentiation yields

$$\frac{Q'(t)}{Q(t)} = \frac{\frac{1}{4}\frac{\cosh\frac{t}{2}}{\sinh^2\frac{t}{2}} - \frac{1}{t^2}}{\frac{1}{t} - \frac{1}{2\sinh\frac{t}{2}}} = \frac{t^2\cosh\frac{t}{2} - 4\sinh^2\frac{t}{2}}{4t\sinh^2\frac{t}{2} - 2t^2\sinh\frac{t}{2}},$$

$$\begin{aligned}\left(\frac{Q'(t)}{Q(t)}\right)' &= \frac{1}{4t^2\left(t-2\sinh\frac{t}{2}\right)^2\sinh^2\frac{t}{2}}(-16t\sinh^3\frac{t}{2} + t^4\cosh^2\frac{t}{2} + 2t^3\sinh^3\frac{t}{2} \\ &\quad -t^4\sinh^2\frac{t}{2} + 16\sinh^4\frac{t}{2} + 8t^2\cosh\frac{t}{2}\sinh^2\frac{t}{2} - 4t^3\cosh^2\frac{t}{2}\sinh\frac{t}{2}) \\ &:= \frac{p\left(\frac{t}{2}\right)}{4t^2\left(t-2\sinh\frac{t}{2}\right)^2\sinh^2\frac{t}{2}},\end{aligned}$$

where

$$\begin{aligned}p(t) &= 16(t^4\cosh^2 t + t^3\sinh^3 t - t^4\sinh^2 t + \sinh^4 t - 2t\sinh^3 t \\ &\quad +2t^2\cosh t\sinh^2 t - 2t^3\cosh^2 t\sinh t.\end{aligned}$$

If we prove that there is a unique number $t_0 > 0$ such that Q'/Q is decreasing on $(0, t_0)$ and increasing on (t_0, ∞), then we have $Q'(t)/Q(t) \geq Q'(t_0)/Q(t_0)$ for all $t > 0$. For this end, we use the "product into sum" formulas and Taylor expansion to get

$$\begin{aligned}\frac{1}{2}p(t) &= \cosh 4t + 4t^2\cosh 3t - 2t^3\sinh 3t - 4t\sinh 3t - 4\cosh 2t \\ &\quad -4t^2\cosh t - 10t^3\sinh t + 12t\sinh t + 8t^4 + 3 \\ &= \sum_{n=0}^{\infty}\frac{4^{2n}t^{2n}}{(2n)!} + 4\sum_{n=1}^{\infty}\frac{3^{2n-2}t^{2n}}{(2n-2)!} - 2\sum_{n=2}^{\infty}\frac{3^{2n-3}t^{2n}}{(2n-3)!} - 4\sum_{n=1}^{\infty}\frac{3^{2n-1}t^{2n}}{(2n-1)!} \\ &\quad -4\sum_{n=0}^{\infty}\frac{2^{2n}t^{2n}}{(2n)!} - 4\sum_{n=1}^{\infty}\frac{t^{2n}}{(2n-2)!} - 10\sum_{n=2}^{\infty}\frac{t^{2n}}{(2n-3)!} \\ &\quad +12\sum_{n=1}^{\infty}\frac{t^{2n}}{(2n-1)!} + 8t^4 + 3 \\ &:= \sum_{n=3}^{\infty}\frac{u_n}{(2n)!}t^{2n},\end{aligned}$$

where

$$u_n = \left(4^{2n} - 8n\left(2n^2 - 9n + 13\right)3^{2n-3} - 2^{2n+2} - 8n\left(10n^2 - 13n + 1\right)\right).$$

A direct verification gives $u_3 = 0$, $u_n < 0$ for $4 \leq n \leq 10$, and we now show that $u_n \geq 0$ for $n \geq 11$. It is easy to check that u_n satisfies the recursive relation

$$\begin{aligned} u_{n+1} - 16u_n &= 8\left(14n^3 - 117n^2 + 199n - 54\right)3^{2n-3} + 48 \times 2^{2n} \\ &\quad +1200n^3 - 1800n^2 + 88n + 16 \\ &> 0 \text{ for } n \geq 11, \end{aligned}$$

which together with $u_{11} = 1636\,643\,754\,240 > 0$ indicates that $u_n \geq 0$ for $n \geq 11$.

By Lemma 2 we see that there is a unique $t_0' > 0$ such that $p(t) < 0$ for $t \in (0, t_0')$ and $p(t) < 0$ for $t \in (t_0', \infty)$, which implies that Q'/Q is decreasing on $(0, 2t_0')$ and increasing on $(2t_0', \infty)$. Numeric calculation shows that $t_0 = 2t_0' = 15.40151...$, and so $Q'(t_0)/Q(t_0) \approx -0.061875$.

This completes the proof. □

To simplify the proofs of some crucial inequalities, we need to the following lemma.

Lemma 4. *For $t > 0$, the following inequalities hold:*

$$\frac{\sinh t}{t} > 3\frac{2\cosh t + 3}{\cosh t + 14}, \tag{2.8}$$

$$\frac{\sinh t}{t} > 15\frac{2\cosh^2 t + 10\cosh t + 9}{2\cosh^2 t + 101\cosh t + 212}, \tag{2.9}$$

$$\frac{\sinh t}{t} < 15\frac{18\cosh^2 t + 160\cosh t + 179}{1159\cosh^2 t + 4192\cosh t + 4}\cosh t. \tag{2.10}$$

Proof. The inequality (2.8) has been proven in [36, Theorem 18].

To prove (2.9), it suffices to show that for $t > 0$

$$p_1(t) := \frac{2\cosh^2 t + 101\cosh t + 212}{2\cosh^2 t + 10\cosh t + 9}\sinh t - 15t > 0.$$

Differentiating and factoring give

$$\begin{aligned} p_1'(t) &= \tfrac{2\cosh^2 t+101\cosh t+212}{2\cosh^2 t+10\cosh t+9}\cosh t - 7\tfrac{26\cosh^2 t+116\cosh t+173}{(2\cosh^2 t+10\cosh t+9)^2}\sinh^2 t - 15 \\ &= \frac{4(\cosh t - 1)^5}{\left(2\cosh^2 t + 10\cosh t + 9\right)^2} > 0, \end{aligned}$$

which yields $p_1(t) > p_1(0) = 0$.

Similarly, inequality (2.10) is equivalent to

$$p_2(t) := \frac{1159\cosh^2 t + 4192\cosh t + 4}{\left(18\cosh^2 t + 160\cosh t + 179\right)\cosh t}\sinh t - 15t < 0.$$

Differentiating and factoring lead us to

$$\begin{aligned} p_2'(t) &= \tfrac{1159\cosh^2 t+4192\cosh t+4}{(18\cosh^2 t+160\cosh t+179)\cosh t}\cosh t + \left(\sinh^2 t\right)\frac{d}{dx}\left(\tfrac{1159x^2+4192x+4}{(18x^2+160x+179)x}\right) - 15 \\ &= -\frac{4(1215x + 179)(x-1)^5}{x^2\left(18x^2 + 160x + 179\right)^2} < 0, \end{aligned}$$

where $x = \cosh t > 1$. This implies that $p_2(t) < p_2(0) = 0$. □

Lemma 5. *The function Q defined by (2.1) satisfies*

$$q_1(t) := Q''(t) + \frac{7}{40}Q(t) > 0$$

for all $t > 0$.

Proof. Differentiation yields

$$Q''(t) = \frac{1}{8\sinh\frac{t}{2}} - \frac{1}{4}\frac{\cosh^2\frac{t}{2}}{\sinh^3\frac{t}{2}} + \frac{2}{t^3}, \tag{2.11}$$

and then

$$q_1(t) = -\frac{1}{80}\frac{20t^3\cosh^2\frac{t}{2} - 14t^2\sinh^3\frac{t}{2} - 3t^3\sinh^2\frac{t}{2} - 160\sinh^3\frac{t}{2}}{t^3\sinh^3\frac{t}{2}}.$$

To prove the desired inequality, we write $q_1(2t)$ in the form of

$$\left(80\sinh^3 t\right) q_1(2t) = 20\left(\frac{\sinh t}{t}\right)^3 + 7\left(\sinh^2 t\right)\frac{\sinh t}{t} + 3\sinh^2 t - 20\cosh^2 t.$$

Employing inequality (2.8), we get

$$\begin{aligned}\left(80\sinh^3 t\right) q_1(2t) &> 20\left(3\frac{2\cosh t+3}{\cosh t+14}\right)^3 + 7\left(\cosh^2 t - 1\right)\left(3\frac{2\cosh t+3}{\cosh t+14}\right)\\ &\quad +3\left(\cosh^2 t - 1\right) - 20\cosh^2 t\\ &= 25\left(\cosh^2 t + 24\cosh t + 240\right)\frac{(\cosh t - 1)^3}{(\cosh t + 14)^3} > 0,\end{aligned}$$

which proves the desired inequality. □

Lemma 6. *The function Q defined by (2.1) satisfies*

$$q_2(t) := Q^{(4)}(t) - \frac{31}{336}Q(t) < 0$$

for all $t > 0$.

Proof. Differentiation yields

$$Q^{(4)}(t) = \frac{7}{8}\frac{\cosh^2\frac{1}{2}t}{\sinh^3\frac{1}{2}t} - \frac{3}{4}\frac{\cosh^4\frac{1}{2}t}{\sinh^5\frac{1}{2}t} - \frac{5}{32\sinh\frac{1}{2}t} + \frac{24}{t^5}, \tag{2.12}$$

and then

$$q_2(t) = -\frac{1}{336}\frac{252t^5\cosh^4\frac{t}{2} + 31t^4\sinh^5\frac{t}{2} + 37t^5\sinh^4\frac{t}{2} - 8064\sinh^5\frac{t}{2} - 294t^5\cosh^2\frac{t}{2}\sinh^2\frac{t}{2}}{t^5\sinh^5\frac{t}{2}}.$$

A simple identical transformation gives

$$\begin{aligned}-\left(672\sinh^5 t\right) q_2(2t) &= -504\left(\frac{\sinh t}{t}\right)^5 + 31\left(\sinh^4 t\right)\frac{\sinh t}{t}\\ &\quad +504\cosh^4 t - 588\cosh^2 t\sinh^2 t + 74\sinh^4 t.\end{aligned}$$

In order to prove the desired inequality, we define

$$U(y) = -504y^5 + 31(\sinh^4 t)y + 504\cosh^4 t - 588\cosh^2 t\sinh^2 t + 74\sinh^4 t,$$

and it suffices to prove that $U((\sinh t)/t) > 0$ for $t > 0$.

An easy computation yields $U'(y) = 31\sinh^4 t - 2520y^4$, which reveals that U is increasing for $y \in \left(1, \sqrt[4]{31/2520}\sinh t\right)$ and decreasing for $y \in [\sqrt[4]{31/2520}\sinh t, \infty)$. We now distinguish two cases to prove $U((\sinh t)/t) > 0$ for $t > 0$.

In the case when $t \in \left(\sqrt[4]{2520/31}, \infty\right)$, by inequality (2.8), we have

$$1 < 3\frac{2\cosh t + 3}{\cosh t + 14} < \frac{\sinh t}{t} < \sqrt[4]{\frac{31}{2520}}\sinh t,$$

that is,

$$3\frac{2\cosh t + 3}{\cosh t + 14}, \frac{\sinh t}{t} \in \left(1, \sqrt[4]{31/2520}\sinh t\right),$$

and so

$$\begin{aligned} U\left(\frac{\sinh t}{t}\right) &> U\left(3\frac{2\cosh t + 3}{\cosh t + 14}\right) = \left(504\cosh^4 t - 588\cosh^2 t\sinh^2 t + 74\sinh^4 t\right) \\ &\quad +31\left(\sinh^4 t\right) \times 3\frac{2\cosh t + 3}{\cosh t + 14} - 504\left(3\frac{2\cosh t + 3}{\cosh t + 14}\right)^5. \end{aligned}$$

Putting $\cosh t = x$, then $\sinh^2 t = x^2 - 1$, and factoring yield

$$U\left(\frac{\sinh t}{t}\right) > \frac{(x-1)^3}{(x+14)^5}U_1(x)$$

where

$$\begin{aligned} U_1(x) &= 176x^6 + 10\,523x^5 + 245\,869x^4 + 2810\,864x^3 \\ &\quad +12\,467\,224x^2 + 12\,511\,688x - 20\,756\,344. \end{aligned}$$

It is easy to verify that $U_1(x) > U_1(1) = 7290\,000 > 0$, which implies that $U((\sinh t)/t) > 0$ for $t \in \left(\sqrt[4]{2520/31}, \infty\right)$.

In the case of when $t \in (0, \sqrt[4]{2520/31}]$, from inequality (2.10) we see that

$$\infty > 15\frac{18\cosh^2 t + 160\cosh t + 179}{1159\cosh^2 t + 4192\cosh t + 4}\cosh t > \frac{\sinh t}{t} > \sqrt[4]{\frac{31}{2520}}\sinh t.$$

Using the decreasing property of U for $u \in \left(\sqrt[4]{31/2520}\sinh t, \infty\right)$, we have

$$U\left(\frac{\sinh t}{t}\right) > U\left(15\frac{18\cosh^2 t + 160\cosh t + 179}{1159\cosh^2 t + 4192\cosh t + 4}\cosh t\right).$$

Letting $\cosh t = x$ and factoring indicate that

$$U\left(15\frac{18x^2 + 160x + 179}{1159x^2 + 4192x + 4}x\right) = \frac{(x-1)^4}{(1159x^2 + 4192x + 4)^5}U_2(x),$$

where

$$\begin{aligned} U_2(x) &= 14\,379\,675\,269\,523\,570x^{11} + 357\,214\,567\,270\,415\,330x^{10} \\ &\quad +3604\,910\,878\,299\,956\,955x^9 + 19\,027\,526\,850\,473\,930\,600x^8 \\ &\quad +55\,570\,610\,110\,726\,848\,080x^7 + 85\,295\,682\,448\,077\,545\,696x^6 \\ &\quad +54\,079\,668\,524\,631\,977\,864x^5 + 560\,130\,320\,580\,220\,160x^4 \\ &\quad +1016\,873\,963\,329\,280x^3 + 923\,378\,178\,560x^2 + 418\,677\,504x + 75\,776. \end{aligned}$$

Evidently, $U_2(x) > 0$, and so $U((\sinh t)/t) > 0$ for $t \in (0, \sqrt[4]{2520/31}]$.

This proof is completed. □

Lemma 7. *The function Q defined by (2.1) satisfies*

$$q_3(t) := Q^{(4)}(t) + \frac{11\,165}{8284}Q''(t) + \frac{199\,849}{1391\,712}Q(t) > 0$$

for all $t > 0$.

Proof. From (2.1), (2.11) and (2.12) it is obtained that

$$\begin{aligned} q_3(t) &= \tfrac{1}{2783\,424t^5\sinh^5\frac{t}{2}}(-2087\,568t^5\cosh^4\frac{t}{2} + 1497\,636t^5\cosh^2\frac{t}{2}\sinh^2\frac{t}{2} \\ &\quad -165\,829t^5\sinh^4\frac{t}{2} + 399\,698t^4\sinh^5\frac{t}{2} \\ &\quad +7502\,880t^2\sinh^5\frac{t}{2} + 66\,802\,176\sinh^5\frac{t}{2}). \end{aligned}$$

We write $q_3(2t)$ as

$$\begin{aligned} &\left(2783\,424\sinh^5 t\right) q_3(2t) \\ = &-\left(2087\,568\cosh^4 t + 165\,829\sinh^4 t - 1497\,636\cosh^2 t\sinh^2 t\right) \\ &+199\,849\left(\sinh^4 t\right)\tfrac{\sinh t}{t} + 937\,860\left(\sinh^2 t\right)\left(\tfrac{\sinh t}{t}\right)^3 + 2087\,568\left(\tfrac{\sinh t}{t}\right)^5. \end{aligned}$$

Utilizing inequality (2.9) and putting $\cosh t = x > 1$ and then factoring yield

$$\begin{aligned} \left(2783\,424\sinh^5 t\right) q_3(2t) &> -\left(2087\,568x^4 + 165\,829\left(x^2-1\right)^2 - 1497\,636x^2\left(x^2-1\right)\right) \\ &\quad +199\,849\left(x^2-1\right)^2\left(15\frac{2x^2+10x+9}{2x^2+101x+212}\right) \\ &\quad +937\,860\left(x^2-1\right)\left(15\frac{2x^2+10x+9}{2x^2+101x+212}\right)^3 \\ &\quad +2087\,568\left(15\frac{2x^2+10x+9}{2x^2+101x+212}\right)^5 \\ &= \frac{7(x-1)^5}{\left(2x^2+101x+212\right)^5} q_4(x) > 0, \end{aligned}$$

where the last inequality holds due to

$$\begin{aligned} q_4(x) &= 10\,249\,024x^9 + 2015\,594\,800x^8 + 163\,876\,520\,192x^7 + 6681\,271\,280\,040x^6 \\ &\quad +136\,012\,433\,414\,956x^5 + 1069\,481\,086\,377\,851x^4 + 4121\,483\,475\,973\,500x^3 \\ &\quad +8450\,810\,874\,059\,188x^2 + 8899\,895\,239\,232\,240x + 3802\,278\,457\,617\,584. \end{aligned}$$

This completes the proof. □

3. Main Results

Now we state and prove the first result, which shows that the second conjecture posed by Chen is valid.

Theorem 1. *Let the function R be defined on $(0,\infty)$ by (1.10). Then the function*

$$h_a(x) = (x+a)^2 R(x)$$

is strictly completely monotonic on $(0,\infty)$ if $a \geq a_0 = \sqrt{c_0^2 + 7/40} - c_0 \approx 0.48476$, where $c_0 \approx -0.061875$ is defined by (2.7).

Proof. Using the relations (2.2)–(2.4) we get that

$$\begin{aligned} h_a(x) &= (x+a)^2 R(x) = x^2 R(x) + 2axR(x) + a^2 R(x) \\ &= \frac{1}{24} + \int_0^\infty e^{-xt} Q''(t)\,dt + 2a\int_0^\infty e^{-xt} Q'(t)\,dt + a^2 \int_0^\infty e^{-xt} Q(t)\,dt \\ &= \frac{1}{24} + \int_0^\infty e^{-xt}\left(Q''(t) + 2aQ'(t) + a^2 Q(t)\right) dt \\ &\triangleq \frac{1}{24} + \int_0^\infty e^{-xt} Q(t)\,\delta_a(t)\,dt. \end{aligned}$$

Evidently, for $t > 0$, $Q(t) > 0$ and by Lemmas 3 and 5,

$$\begin{aligned} \delta_a(t) &= \frac{Q''(t)}{Q(t)} + 2a\frac{Q'(t)}{Q(t)} + a^2 \geq a^2 + 2ac_0 - \frac{7}{40} \\ &= \left(a + c_0 + \sqrt{c_0^2 + \frac{7}{40}}\right)\left(a + c_0 - \sqrt{c_0^2 + \frac{7}{40}}\right) \geq 0 \end{aligned}$$

if $a \geq \sqrt{c_0^2 + 7/40} - c_0$, which proves the desired result. □

Taking $a = 1/2$ and replacing x by $(x + 1/2)$ in the above theorem, we have

Corollary 1. *The function* $((x+1)/x)^2 H(x)$ *is strictly completely monotonic on* $(-1/2, \infty)$.

Remark 1. *Evidently, Theorem 1 reveals that the second conjecture posed by Chen in* [15, Theorem 2] *is true.*

Our second result states that

Theorem 2. *Let the function* R *be defined on* $(0, \infty)$ *by (1.10). Then the function*

$$x \mapsto F_a(x) = 24\left(x^2 + a\right) R(x) - 1$$

is strictly completely monotonic on $(0, \infty)$ *if and only if* $a \geq a_1 = 17/40$.

Proof. The necessity follows from

$$\lim_{x\to\infty} \frac{F_a(x)}{x^{-2}} = \lim_{x\to\infty} \frac{24\left(x^2 + a\right)\left(\psi(x + 1/2) - \ln x\right) - 1}{x^{-2}} = a - \frac{7}{40} \geq 0.$$

Using the relations (2.2) and (2.4) we obtain that

$$\begin{aligned} F_a(x) &= 24\left(x^2 + a\right) R(x) - 1 = 24x^2 R(x) + 24aR(x) - 1 \\ &= 24\left(\frac{1}{24} + \int_0^\infty e^{-xt} Q''(t)\,dt\right) + 24a\int_0^\infty e^{-xt} Q(t)\,dt - 1 \\ &= 24\int_0^\infty e^{-xt}\left(Q''(t) + aQ(t)\right) dt. \end{aligned}$$

If $a \geq 7/40$, then by Lemma 5 it follows that

$$Q''(t) + aQ(t) \geq Q''(t) + \frac{7}{40} Q(t) > 0$$

for $t > 0$, which proves the sufficiency. □

Noting that

$$F_{7/40}\left(n+\frac{1}{2}\right)=24\left(\left(n+1/2\right)^2+7/40\right)R_n-1$$

and the facts that

$$F_{7/40}\left(3/2\right)=\frac{286}{5}-\frac{291}{5}\ln\frac{3}{2}-\frac{291}{5}\gamma\approx 0.007979 \text{ and } F_{7/40}\left(\infty\right)=0,$$

by Theorem 3 we get immediately

Corollary 2. *Let R_n be defined by (1.3). Then the double inequality*

$$\frac{1}{24\left(\left(n+1/2\right)^2+7/40\right)}<R_n-\gamma<\frac{1+\lambda_1}{24\left(\left(n+1/2\right)^2+7/40\right)} \tag{3.1}$$

holds for $n\in\mathbb{N}$, where $\lambda_1=f_{7/40}\left(3/2\right)\approx 0.007979$ is the best constant.

Our third result is contained in the following theorem.

Theorem 3. *Let the function R be defined on $(0,\infty)$ by (1.10). Then the function*

$$x\mapsto f_a\left(x\right)=-24\left(x^4+a\right)R\left(x\right)+x^2-\frac{7}{40},$$

is strictly completely monotonic on $(0,\infty)$ is and only if $a\le a_2=-31/336$.

Proof. The necessity can be deduced by

$$\lim_{x\to\infty}\frac{f_a\left(x\right)}{x^{-2}}=\lim_{x\to\infty}\frac{-24\left(x^4+a\right)\left(\psi\left(x+1/2\right)-\ln x\right)+x^2-\frac{7}{40}}{x^{-2}}=-a-\frac{31}{336}\ge 0.$$

By the relations (2.2), (2.4) and (2.6), we obtain that

$$\begin{aligned}
f_a\left(x\right) &= -24x^4R\left(x\right)-24aR\left(x\right)+x^2-\frac{7}{40}\\
&= -24\left(\frac{1}{24}x^2-\frac{7}{960}+\int_0^\infty e^{-xt}Q^{(4)}\left(t\right)dt\right)-24a\int_0^\infty e^{-xt}Q\left(t\right)dt+x^2-\frac{7}{40}\\
&= 24\int_0^\infty e^{-xt}\left(-Q^{(4)}\left(t\right)-aQ\left(t\right)\right)dt.
\end{aligned}$$

If $a\le a_2=-31/336$, then by Lemma 6 it follows that

$$-Q^{(4)}\left(t\right)-aQ\left(t\right)\ge -Q^{(4)}\left(t\right)+\frac{31}{336}Q\left(t\right)>0,$$

which proves the sufficiency. □

Utilizing the decreasing property of f_{a_2} and noting the facts that

$$f_{a_2}\left(\frac{3}{2}\right)=\frac{835}{7}\gamma+\frac{835}{7}\ln\frac{3}{2}-\frac{32\,819}{280}\approx 0.0090636 \text{ and } f_{a_2}\left(\infty\right)=0,$$

we have

Corollary 3. *Let R_n be defined by (1.3). Then the double inequality*

$$\frac{1}{24}\frac{\left(n+\frac{1}{2}\right)^2-\frac{7}{40}-\lambda_2}{\left(n+\frac{1}{2}\right)^4-\frac{31}{336}}<R_n-\gamma<\frac{1}{24}\frac{\left(n+\frac{1}{2}\right)^2-\frac{7}{40}}{\left(n+\frac{1}{2}\right)^4-\frac{31}{336}} \tag{3.2}$$

holds for $n\in\mathbb{N}$, where $\lambda_2=f_{a_2}\left(3/2\right)\approx 0.0090636$ is the best constant.

Remark 2. *The upper bound for $R_n-\gamma$ given in (3.2) is better than the one given in (1.5), because*

$$\begin{aligned}&\frac{1}{24}\frac{\left(n+\frac{1}{2}\right)^2-\frac{7}{40}}{\left(n+\frac{1}{2}\right)^4-\frac{31}{336}}-\left(\frac{1}{24n^2}+\frac{7}{960}\frac{1}{n^4}\right)\\ =\;&-\frac{6720n^5+10\,752n^4+5712n^3+1564n^2+588n-35}{960n^4\left(168n^4+336n^3+252n^2+84n-5\right)}<0.\end{aligned}$$

The last result is the following theorem

Theorem 4. *Let the function R be defined on $(0,\infty)$ by (1.10). Then the function*

$$x\mapsto G_a(x)=24\left(x^4+ax^2+\frac{7}{40}a-\frac{31}{336}\right)R(x)-\left(x^2-\frac{7}{40}+a\right) \tag{3.3}$$

is strictly completely monotonic on $(0,\infty)$ if and only if $a\geq a_3=11165/8284$.

Proof. The necessity can be deduced by

$$\begin{aligned}\lim_{x\to\infty}\frac{G_a(x)}{x^{-4}}&=\lim_{x\to\infty}\frac{24\left(x^4+ax^2+\frac{7}{40}a-\frac{31}{336}\right)(\operatorname{Psi}(x+1/2)-\ln x)-\left(x^2-\frac{7}{40}+a\right)}{x^{-4}}\\ &=\frac{2071}{33\,600}\left(a-\frac{11\,165}{8284}\right)\geq 0.\end{aligned}$$

By the relations (2.2), (2.4) and (2.6) we obtain that

$$\begin{aligned}G_a(x)&=24x^4R(x)+24ax^2R(x)+24\left(\frac{7}{40}a-\frac{31}{336}\right)R(x)-\left(x^2-\frac{7}{40}+a\right)\\ &=24\left(\frac{1}{24}x^2-\frac{7}{960}+\int_0^\infty e^{-xt}Q^{(4)}(t)\,dt\right)+24a\left(\frac{1}{24}+\int_0^\infty e^{-xt}Q''(t)\,dt\right)\\ &\quad+24\left(\frac{7}{40}a-\frac{31}{336}\right)\int_0^\infty e^{-xt}Q(t)\,dt-\left(x^2-\frac{7}{40}+a\right)\\ &=24\int_0^\infty e^{-xt}\left(Q^{(4)}(t)+aQ''(t)+\left(\frac{7}{40}a-\frac{31}{336}\right)Q(t)\right)dt\\ &\triangleq 24\int_0^\infty e^{-xt}g_a(t)\,dt.\end{aligned}$$

If $a\geq a_3=11165/8284$, then it follows from Lemmas 5 and 7 that

$$\begin{aligned}g_a(t)&=Q^{(4)}(t)-\frac{31}{336}Q(t)+a\left(Q''(t)+\frac{7}{40}Q(t)\right)\\ &\geq Q^{(4)}(t)-\frac{31}{336}Q(t)+\frac{11\,165}{8284}\left(Q''(t)+\frac{7}{40}Q(t)\right)\\ &=Q^{(4)}(t)+\frac{11\,165}{8284}Q''(t)+\frac{199\,849}{1391\,712}Q(t)>0,\end{aligned}$$

which proves the sufficiency. □

Application of the decreasing property of G_{a_3} and notice the facts that

$$G_{a_3}\left(\frac{3}{2}\right)=\frac{112\,672\,809}{579\,880}-\frac{11\,465\,761}{57\,988}\ln\frac{3}{2}-\frac{11\,465\,761}{57\,988}\gamma\approx 0.0016903 \text{ and } G_{a_3}(\infty)=0,$$

we have

Corollary 4. *Let R_n be defined by (1.3). Then the double inequality*

$$\frac{1}{24}\frac{\left(n+\frac{1}{2}\right)^2+\frac{97\,153}{82\,840}}{\left(n+\frac{1}{2}\right)^4+\frac{11\,165}{8284}\left(n+\frac{1}{2}\right)^2+\frac{199\,849}{1391\,712}}<R_n-\gamma<\frac{1}{24}\frac{\left(n+\frac{1}{2}\right)^2+\frac{97\,153}{82\,840}+\lambda_3}{\left(n+\frac{1}{2}\right)^4+\frac{11\,165}{8284}\left(n+\frac{1}{2}\right)^2+\frac{199\,849}{1391\,712}} \tag{3.4}$$

holds for $n\in\mathbb{N}$, where $\lambda_3=G_{a_3}(3/2)\approx 0.0016903$ is the best constant.

4. Remarks

Remark 3. *The function G_a defined by (3.3) can be written as*

$$G_a(x)=a\times f_{7/40}(x)-F_{-31/336}(x)=f_{7/40}(x)\times\left(a-\frac{F_{-31/336}(x)}{f_{7/40}(x)}\right). \tag{4.1}$$

Theorem 4 tell us that

$$\frac{F_{-31/336}(x)}{f_{7/40}(x)}\le\lim_{x\to\infty}\frac{F_{-31/336}(x)}{f_{7/40}(x)}=\lim_{x\to\infty}\frac{-24\left(x^4-\frac{31}{336}\right)R(x)+x^2-\frac{7}{40}}{24\left(x^2+\frac{7}{40}\right)R(x)-1}\le\frac{11\,165}{8284}. \tag{4.2}$$

On the other hand, we can prove that

$$\frac{F_{-31/336}(x)}{f_{7/40}(x)}\ge\lim_{x\to0^+}\frac{F_{-31/336}(x)}{f_{7/40}(x)}=\lim_{x\to0^+}\frac{-24\left(x^4-\frac{31}{336}\right)R(x)+x^2-\frac{7}{40}}{24\left(x^2+\frac{7}{40}\right)R(x)-1}\ge\frac{155}{294}. \tag{4.3}$$

It suffices to prove the function

$$x\mapsto V(x)=\psi(x+1/2)-\ln x-\frac{1}{24}\frac{x^2+\frac{2071}{5880}}{x^2\left(x^2+\frac{155}{294}\right)}$$

is increasing on $(0,\infty)$. Differentiation gives

$$V'(x)=\psi'(x+1/2)-\frac{1}{x}+\frac{1728\,720x^4+1217\,748x^2+321\,005}{10x^3\left(294x^2+155\right)^2}.$$

Utilizing $\psi'(x+1)-\psi'(x)=-1/x^2$ yields

$$\begin{aligned}V'(x+1)-V'(x)&=-\tfrac{1}{(x+1/2)^2}-\tfrac{1}{x+1}+\tfrac{1728\,720(x+1)^4+1217\,748(x+1)^2+321\,005}{10(x+1)^3\left(294(x+1)^2+155\right)^2}\\&\quad+\tfrac{1}{x}-\tfrac{1728\,720x^4+1217\,748x^2+321\,005}{10x^3(294x^2+155)^2}\\&=-\tfrac{V_1(x)}{10x^3(2x+1)^2(294x^2+155)^2(x+1)^3(294x^2+588x+449)^2},\end{aligned}$$

where

$$\begin{aligned}V_1(x)&=1718\,371\,882\,080x^{12}+10\,310\,231\,292\,480x^{11}+29\,399\,355\,669\,600x^{10}\\&\quad+52\,486\,324\,833\,600x^9+66\,690\,983\,696\,400x^8+65\,258\,530\,001\,280x^7\\&\quad+51\,909\,045\,513\,612x^6+34\,352\,301\,620\,196x^5+18\,881\,999\,450\,054x^4\\&\quad+8378\,736\,976\,048x^3+2808\,871\,359\,013x^2+622\,502\,847\,155x+64\,714\,929\,005.\end{aligned}$$

It is evident that $V_1(x)>0$ for $x>0$, and so $V'(x+1)-V'(x)<0$ for $x>0$. This leads us to

$$V'(x)>V'(x+1)>V'(x+2)>\cdots>\lim_{n\to\infty}V'(x+n)=0,$$

which reveals that V is increasing on $(0,\infty)$.

Meanwhile, (4.1) implies that an necessary condition such that the function $-G_a$ is completely monotone on $(0,\infty)$ is

$$a\le\lim_{x\to0+}\frac{F_{-31/336}(x)}{f_{7/40}(x)}=\lim_{x\to0+}\frac{-24\left(x^4-\frac{31}{336}\right)R(x)+x^2-\frac{7}{40}}{24\left(x^2+\frac{7}{40}\right)R(x)-1}=\frac{155}{294}.$$

This together with inequalities (4.2) and (4.3) yield two conjectures.

Conjecture 1. *Let the function R be defined on $(0,\infty)$ by (1.10). Then*

(i) the function

$$x \mapsto \frac{-24\left(x^4-\frac{31}{336}\right)R(x)+x^2-\frac{7}{40}}{24\left(x^2+\frac{7}{40}\right)R(x)-1}$$

is increasing on $(0,\infty)$;

(ii) the function $-G_a$ is completely monotone on $(0,\infty)$ if and only if $a \leq 155/294$.

Remark 4. *In addition, using the increasing property of the function V proved in Remark 3, and noting that*

$$V\left(\frac{3}{2}\right)=\frac{866\,519}{881\,820}-\ln\frac{3}{2}-\gamma \approx -0.000032387 \text{ and } V(\infty)=0$$

we have

$$\frac{1}{24}\frac{\left(n+\frac{1}{2}\right)^2+\frac{2071}{5880}}{\left(n+\frac{1}{2}\right)^2\left(\left(n+\frac{1}{2}\right)^2+\frac{155}{294}\right)}+\lambda_4 < R_n-\gamma < \frac{1}{24}\frac{\left(n+\frac{1}{2}\right)^2+\frac{2071}{5880}}{\left(n+\frac{1}{2}\right)^2\left(\left(n+\frac{1}{2}\right)^2+\frac{155}{294}\right)},$$

where $\lambda_4 \approx -0.000032387$ is the best possible.

Clearly, it is an improvement of (3.2), since

$$\begin{aligned}&\frac{1}{24}\frac{\left(n+\frac{1}{2}\right)^2+\frac{2071}{5880}}{\left(n+\frac{1}{2}\right)^2\left(\left(n+\frac{1}{2}\right)^2+\frac{155}{294}\right)}-\frac{1}{24}\frac{\left(n+\frac{1}{2}\right)^2-\frac{7}{40}}{\left(n+\frac{1}{2}\right)^4-\frac{31}{336}}\\ =\;& -\frac{64\,201}{120\left(2n+1\right)^2\left(588n^2+588n+457\right)\left(168n^4+336n^3+252n^2+84n-5\right)}<0.\end{aligned}$$

Remark 5. *Theorems 3 and 4 and Remark 3 offer in fact three new sequences convergent to γ with increasingly higher speed. Denote by*

$$\begin{aligned}w_n &= \sum_{k=1}^{n}\frac{1}{k}-\ln\left(n+1/2\right)-\frac{1}{24}\frac{\left(n+\frac{1}{2}\right)^2-\frac{7}{40}}{\left(n+\frac{1}{2}\right)^4-\frac{31}{336}},\\ y_n &= \sum_{k=1}^{n}\frac{1}{k}-\ln\left(n+1/2\right)-\frac{1}{24}\frac{\left(n+\frac{1}{2}\right)^2+\frac{97\,153}{82\,840}}{\left(n+\frac{1}{2}\right)^4+\frac{11\,165}{8284}\left(n+\frac{1}{2}\right)^2+\frac{199\,849}{1391\,712}},\\ z_n &= \sum_{k=1}^{n}\frac{1}{k}-\ln\left(n+1/2\right)-\frac{1}{24}\frac{\left(n+\frac{1}{2}\right)^2+\frac{2071}{5880}}{\left(n+\frac{1}{2}\right)^2\left(\left(n+\frac{1}{2}\right)^2+\frac{155}{294}\right)}\end{aligned}$$

Then by Theorems 3 and 4 and Remark 4 we have

$$w_n < z_n < \gamma < y_n.$$

And, a simple check yields

$$\begin{aligned}\lim_{n\to\infty} n^8\left(w_n-\gamma\right) &= -\frac{319}{92\,160},\\ \lim_{n\to\infty} n^{10}\left(y_n-\gamma\right) &= \frac{627\,404\,761}{246\,900\,842\,496},\\ \lim_{n\to\infty} n^8\left(z_n-\gamma\right) &= -\frac{199\,849}{94\,832\,640}\end{aligned}$$

These show that the sequences (w_n), (y_n) and (z_n) converge to γ as n^{-8}, n^{-10} and n^{-8}, respectively.

Lastly, inspired by Theorem 2–4, we post the following open problem.

Problem 1. *Determine the best constants a_k and b_k such that the function*

$$x \mapsto \left(\sum_{k=0}^{n+1} a_k x^{2k}\right) R(x) - \left(\sum_{k=0}^{n} b_k x^{2k}\right)$$

is completely monotone on $(0,\infty)$, and satisfies that

$$\lim_{x\to\infty} \frac{\left(\sum_{k=0}^{n+1} a_k x^{2k}\right) R(x) - \left(\sum_{k=0}^{n} b_k x^{2k}\right)}{x^{-2n-4}} = c \neq 0, \pm\infty.$$

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